

MPHARM

(SEM. II) THEORY EXAMINATION 2025-26

PRINCIPLES OF DRUG DISCOVERY

TIME: 3 HRS

M.MARKS: 75

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

10x2 = 20

- (a) Define zinc finger proteins with suitable example.
- (b) List and explain the role of genomics in validation studies.
- (c) Summarize the role of combinatorial chemistry in lead discovery techniques?
- (d) Distinguish between protein domains and motifs.
- (e) Write the importance of pharmacophore-based screening in rational drug design.
- (f) Briefly relate Hansch analysis and Free-Wilson analysis.
- (g) How molecular docking and QSAR complement each other in rational drug design.
- (h) State the relationship between Hansch and Free-Wilson analysis.
- (i) What is regression analysis in QSAR studies?
- (j) How the prodrugs improve sustained drug action with suitable example?

SECTION B

2. Attempt any two of the following:

2 x 10 = 20

- (a) Discuss in detail about lead identification and optimization for discovery of new molecule.
- (b) Describe the steps involved docking. Explain in detail with suitable example.
- (c) Explain the role of statistical analysis used in QSAR. Describe in detail about partial least square method.

SECTION C

3. Attempt any five part of the following:

7 x 5 = 35

- (a) Define microarray. Explain in detail about protein microarrays in the modern drug discovery process.
- (b) Explain assay development for hit identification and the key steps involved in the drug discovery process.
- (c) Briefly describe high-throughput screening with an application.
- (d) Explain in different physicochemical parameters involved in QSAR.
- (e) Explain how prodrugs to improve drug absorption and distribution with suitable example.
- (f) Describe the concept of homology modelling. Explain in detail, with its limitations in modern drug discovery.
- (g) Differentiate between COMFA and COMSIA with example.